TEMPLE COLLEGE
WORKFORCE DEVELOPMENT

CHIEFOS

STUDENT REVIEW PACKET

Week 1

IDE concept, VS Code tour, terminal, extensions

Phase 1 — Foundations · Weeks 1–6 · Modules 00–02

No D2L login required. This is the same lesson text posted in Brightspace for this week. Work the lessons in order on your own machine. Keep each checkpoint screenshot or pasted output. Bring it to the live session, or email it to your instructor, instead of submitting in D2L. Module quizzes stay in D2L and are not included here.

In this packet

1. Module 00 — VS Code Walkthrough: overview
2. 00.00 What is an IDE?
3. 00.01 VS Code UI Tour
4. 00.02 The Integrated Terminal
5. 00.03 Extensions Overview
6. 00.04 The Docker Extension

Module 00 — VS Code Walkthrough

What you'll be able to do after this module

- Explain what an IDE is and how VS Code combines an editor, a terminal, and other tools in one window
- Navigate the VS Code UI, naming the activity bar, side bar, editor, panel, and status bar, and run commands from the command palette
- Open the integrated terminal and distinguish typing into the terminal from typing into the editor
- Install, disable, and uninstall extensions from the marketplace, verifying the publisher before installing
- Install the Docker extension and explain why its sidebar is empty before Docker itself is installed
- Connect VS Code to WSL on Windows and open a folder in the Linux home directory
- Connect VS Code to a remote machine over SSH, open a remote folder, and verify the terminal runs on the remote

Lessons and objectives

#	Lesson	Learning objective
00	What is an IDE?	Install VS Code and explain the difference between an editor, a terminal, and an IDE.
01	VS Code UI Tour	Identify the activity bar, side bar, editor, panel, and status bar, and use the command palette (`Ctrl+Shift+P`) to toggle the terminal and change the editor font size.
02	The Integrated Terminal	Open the integrated terminal, run `echo` commands at the prompt, and distinguish typing into the terminal from typing into the editor.
03	Extensions Overview	Install the Python extension by Microsoft from the marketplace, verify its publisher first, and describe when to use Install, Disable, and Uninstall.
04	The Docker Extension	Install the Docker extension by Microsoft, locate its whale icon and sidebar sections, and explain why the sidebar shows 'Failed to connect' before Docker is installed.
05	Connecting to WSL	Connect VS Code to WSL using the WSL extension, open a folder in the Linux home directory, and verify the terminal is Linux by running `uname -a`.
06	Connecting over SSH	Connect VS Code to an instructor-provided remote host with the Remote - SSH extension, open a remote folder, and confirm the session by running `hostname` and `whoami`.
07	Final Checkpoint	Demonstrate unaided use of VS Code by toggling the terminal from the command palette, installing an extension, connecting to WSL or SSH, and capturing a screenshot showing the green remote indicator with `whoami` and `pwd` output.

Each lesson is one learning object with a paired lab. Work in order — completing a lab unlocks the next lesson. The module ends with a **Check on Learning** quiz and a hands-on final checkpoint.

Learning objective: Install VS Code and explain the difference between an editor, a terminal, and an IDE.

Lesson 00: What is an IDE?

Time: ~15 min

Concepts

Before we install anything, three words you'll hear all day:

- **Editor** — a program that lets you type and save text into a file. Notepad on Windows, TextEdit on Mac. That's it. No magic.
- **Terminal** — a program where you type commands as text and the computer runs them. No buttons, no menus. Just text in, text out. The Windows "Command Prompt" and the Mac "Terminal app" are both terminals.
- **IDE** — short for **Integrated Development Environment**. It is an editor and a terminal and a bunch of other tools, all bolted together in one window. The whole point of an IDE is that you don't have to alt-tab between five different programs to write, run, and debug code.

VS Code is an IDE. VS Code (Visual Studio Code, made by Microsoft) is one of the most widely used IDEs in the world. Because it's so popular, if you ever google something like "how do I do X in VS Code" the answer will match exactly what's on your screen.

> **A note on AI for this course:** Out of the box, VS Code has no AI assistant — it's just the editor. In Module 05 we'll install **Continue.dev**, a free, open-source extension that gives you the same agentic-coding features (chat, inline edit, autocomplete) powered by the college's shared LLM server. Today we just want the IDE itself. If VS Code offers to "Sign in to sync settings," that's optional cloud sync and unrelated — skip it.

Demo (instructor at the front)

The instructor will show:

1. Notepad (or TextEdit) — just an editor.
2. Command Prompt (or Terminal app) — just a terminal.
3. VS Code — both, plus more, in one window.

You will see how the same task ("write a file, then run it") looks in all three.

Hands-on: install VS Code

1. Open your normal web browser.
2. Go to **code.visualstudio.com**.
3. Click **Download** for your operating system (Windows / Mac / Linux).
4. Run the installer. Accept the defaults.
5. Launch VS Code.

When VS Code first opens you'll land on a **Welcome** tab. It may offer to:

- **Choose a color theme.** Pick whichever you like. Light or dark — your eyes, your call.
- **Sign in to sync settings.** This is optional cloud sync of your settings across machines. **Skip it for the whole course** — you don't need it.
- **Get Started walkthroughs.** VS Code shows a few "getting started" tips on the Welcome tab. Feel free to close them — we'll cover everything you need as we go.

The shortcuts in the rest of this module assume VS Code's **default** keybindings, which is what you get out of the box. You don't need to change anything.

Checkpoint

You're done with this lesson when:

- [ ] VS Code is installed on your laptop.
- [ ] You have VS Code open and you see a welcome screen (or an empty window with a sidebar on the left).
- [ ] You did **not** need to sign in to anything — VS Code is free and works immediately.
- [ ] You can explain in one sentence what an IDE is. Try it on the person next to you.

If VS Code won't install or won't open, raise your hand. **Don't move on until your laptop is at the checkpoint above.** Every later lesson assumes you're sitting in front of a working VS Code window.

Lab

Objective: Explain editor vs terminal vs IDE, and have VS Code installed and open.

Task: Work through this lesson — install VS Code from code.visualstudio.com, launch it, and land on the Welcome tab. Skip "Sign in to sync settings."

Submit: One screenshot of your full VS Code window showing the Welcome tab (or an empty window with the left sidebar), plus one sentence in the submission comment: what an IDE is, in your own words.

Pass criteria:

- [ ] Screenshot shows VS Code open on your machine.
- [ ] You did not sign in to anything.
- [ ] Your one-sentence definition says an IDE combines an editor, a terminal, and other tools in one window.

Unlocks: 01-vscode-tour.md

Lab — do it for real. Complete every step above on your own machine — type the commands, don't copy-paste. **Submit:** a screenshot or pasted output of the *Checkpoint* section (or this lesson's final verification steps) Keep it. If you cannot submit in D2L, bring it to class or email it to your instructor. Completing the lab unlocks the next lesson.

Learning objective: Identify the activity bar, side bar, editor, panel, and status bar, and use the command palette (`Ctrl+Shift+P`) to toggle the terminal and change the editor font size.

Lesson 01: VS Code UI Tour

Time: ~15 min

> Mac users: every `Ctrl+...` shortcut in this module is `Cmd+...` on your machine. The lesson uses `Ctrl` everywhere — translate as you go.

What you'll learn

The names of every part of the VS Code window. Once you know the names, you can google your way out of any problem.

The parts of the window



- **Activity bar** — the thin strip of icons on the very left edge of the window.
- **Side bar** — the wider panel that opens when you click an activity-bar icon.

- **Editor** — the big middle area where files open. Each open file is a **tab** at the top.
- **Panel** — a horizontal area at the bottom of the editor. The integrated terminal lives here. Hidden by default.
- **Status bar** — the colored strip at the very bottom. The bottom-left section is important — it tells you where VS Code is "connected to" (we'll see this in lessons 05 and 06).

Demo: the activity bar, top to bottom

1. **Files** (stacked icon) — your folder's file tree. The default view.
2. **Search** (magnifying glass) — search across every file in the folder at once.
3. **Source Control** (a Y-shape) — git stuff. We'll come back to this in a later module.
4. **Run and Debug** (a play button with a bug) — run code with breakpoints. Later module.
5. **Extensions** (four squares with one floating) — the marketplace. Lesson 03.
6. **Continue** (added in Module 05) — once we install the Continue.dev extension, its icon appears here for AI chat. It's not installed yet, so you won't see it today.

There may be more or fewer icons depending on your VS Code version and installed extensions. That's fine.

Demo: open a folder

1. **File → Open Folder…** (or `Ctrl+K Ctrl+O`).
2. Pick any folder on your computer — your Desktop, your Documents, anything.
3. The first time VS Code opens that folder it asks "Do you trust the authors of the files in this folder?" Click **Yes**.
4. Look at the file explorer in the side bar. You should see your folder's contents.

Click any file. It opens in the editor as a new tab. You can have many tabs open at once. Click the **×** on a tab to close it.

Demo: the command palette

This is the most important shortcut in the entire IDE.

1. Press **Ctrl+Shift+P** (Mac: Cmd+Shift+P).
2. A search box appears at the top of the window. This is the **command palette**.
3. Type "toggle terminal" — you'll see a match. Press Enter.
4. The integrated terminal opens at the bottom of the editor. (We'll use this in the next lesson.)
5. Press **Ctrl+Shift+P** again, type "toggle terminal" again, Enter — terminal closes.

Anything you can do in VS Code is also available through the command palette by name. If you forget a shortcut, this is how you find it.

Demo: settings

1. Press **Ctrl+,** (comma) — the settings page opens.

2. There's a search box at the top. Type "font size."
3. The first result is "Editor: Font Size." Change it from 14 to **16**. The change is instant — look at any open file, the text is bigger.

Settings save automatically. Close the settings tab when you're done.

Try it yourself

1. Close any folder you have open (`File → Close Folder`).
2. Open a folder of your choosing.
3. Open any file.
4. Use the command palette to toggle the terminal. Then toggle it again to close it.
5. Open settings and change "Editor: Font Size" to **16**.

Checkpoint

You're done when:

- [ ] You can name (out loud) the activity bar, side bar, editor, panel, and status bar.
- [ ] You opened a folder and a file.
- [ ] You opened the command palette with `Ctrl+Shift+P` and ran "Toggle Terminal."
- [ ] Your editor font size is 16.

If you didn't get there, ask a neighbor before raising your hand.

Lab

Objective: Name the five regions of the VS Code window and drive it with the command palette.

Task: Work through this lesson's demos — open a folder and a file, use the command palette (`Ctrl+Shift+P`) to run "Toggle Terminal," and change "Editor: Font Size" to 16 in settings.

Submit: One screenshot of your full VS Code window with your folder open in the side bar and the Settings tab showing "Editor: Font Size" set to 16, plus the five region names (activity bar, side bar, editor, panel, status bar) typed in the submission comment.

Pass criteria:

- [ ] Screenshot shows a folder open in the file explorer.
- [ ] The Settings tab shows "Editor: Font Size" at 16.
- [ ] Your comment names all five regions of the window.

Unlocks: 02-integrated-terminal.md

Lab — do it for real. Complete every step above on your own machine — type the commands, don't copy-paste. **Submit:** a screenshot or pasted output of the *Checkpoint* section (or this lesson's final verification steps) Keep it. If you cannot submit in D2L, bring it to class or email it to your instructor. Completing the lab unlocks the next lesson.

Learning objective: Open the integrated terminal, run `echo` commands at the prompt, and distinguish typing into the terminal from typing into the editor.

Lesson 02: The Integrated Terminal

Time: ~10 min

What is a terminal, again?

A **terminal** is a program where you type a command, press Enter, and the computer runs that command and prints text back at you. That is the entire interface — text in, text out.

You already have a terminal on your computer:

- **Windows:** "Command Prompt" or "PowerShell" or "Windows Terminal."
- **Mac:** "Terminal" (in Applications → Utilities).
- **Linux:** "Terminal" or "Console" (varies by distro).

VS Code has a terminal **built into** its window. That's the **integrated terminal**. It's the same kind of program as the ones above — same commands, same output — just embedded inside the IDE so you don't have to alt-tab to a separate window.

Why this matters

Almost every later module of this course assumes you have a terminal open. **Module 01 is 2.5 hours of typing commands into this terminal.** Today, all you need to know is how to open it and how to type into it.

> We are deliberately **not** teaching you commands like `ls`, `cd`, `pwd`, etc. in this lesson. That's all of Module 01. Today is just "where is the terminal and what does it look like."

Demo: open the integrated terminal

Three ways to open it. Pick whichever you remember:

1. Keyboard shortcut: **`Ctrl+` `** — that's Ctrl + backtick. The backtick is usually on the same key as the tilde (~`), top-left of the keyboard, just under Esc.
2. Menu: **View** → **Terminal**.
3. Command palette: `Ctrl+Shift+P` → type "Toggle Terminal" → Enter.

The terminal opens in a panel at the bottom of the editor. You'll see a **prompt** — something that looks like `user@host:~$` on Linux/Mac or `PS C:\Users\You>` on Windows. The prompt is the terminal saying "I'm

ready, type a command."

Editor vs terminal: don't mix them up

This is the single most common beginner mistake.

- The **editor** is the big area in the middle. When you type there, you're typing **into a file**. Nothing runs.
- The **terminal** is the panel at the bottom. When you type there and press Enter, the computer **runs your command**.

If you type a command into the editor, the editor will just store the text in a file and nothing will happen. If you type a sentence into the terminal, the terminal will try to run it as a command and complain that it doesn't exist.

Always look at where your blinking cursor is before typing.

Try it yourself

1. Open the integrated terminal (any of the three ways above).
2. Click inside the terminal panel — you should see the cursor blinking next to the prompt.
3. Type exactly this and press Enter:

``` `echo hello` ```

You should see `hello` printed on the next line, then a fresh prompt.

1. Now type your own name:

``` `echo your-name-here` ```

(Replace `your-name-here` with your actual name. No quotes needed.)

1. Close the terminal with ``` `Ctrl+` ``` and reopen it. The terminal pane comes back — the previous output is still there.

Checkpoint

You're done when:

- [ ] You opened the integrated terminal at least once.
- [ ] You ran `echo hello` and saw `hello` printed back.
- [ ] You ran `echo` with your own name and it worked.
- [ ] You can explain the difference between typing into the editor and typing into the terminal.

You don't need to memorize any commands today. That's tomorrow's problem (Module 01).

Lab

Objective: Open the integrated terminal and run your first commands in it.

Task: Work through this lesson — open the integrated terminal (`` Ctrl+ `, View → Terminal, or the command palette), then run `echo hello` and `echo` with your own name.

Submit: One screenshot of your VS Code window with the terminal panel showing both `echo` commands and their output.

Pass criteria:

- [ ] The terminal panel is visible at the bottom of the VS Code window (not a separate app).
- [ ] `echo hello` and its `hello` output are visible.
- [ ] `echo` with your own name and its output are visible.

Unlocks: 03-extensions-overview.md

Lab — do it for real. Complete every step above on your own machine — type the commands, don't copy-paste. **Submit:** a screenshot or pasted output of the *Checkpoint* section (or this lesson's final verification steps) Keep it. If you cannot submit in D2L, bring it to class or email it to your instructor. Completing the lab unlocks the next lesson.

Learning objective: Install the Python extension by Microsoft from the marketplace, verify its publisher first, and describe when to use Install, Disable, and Uninstall.

Lesson 03: Extensions Overview

Time: ~15 min

What is an extension?

An **extension** is a plugin that adds features to VS Code. VS Code by itself is a general-purpose IDE. Extensions specialize it: an extension might add Python support, or a Docker sidebar, or a way to connect to a remote computer. You install only the ones you need.

Extensions are made by Microsoft, by community developers, and by companies. The big public list of all extensions is the **marketplace**.

Demo: open the marketplace

1. Click the **Extensions** icon in the activity bar (the four squares with one floating away).

- Or press `Ctrl+Shift+X`.

1. The side bar fills with a search box and a list of popular extensions.

The buttons on an extension

Click any extension in the list. The right-hand pane shows its details: description, version, publisher, screenshots.

The button next to the extension name does one of three things depending on its state:

- **Install** — extension isn't installed yet. Click to install.
- **Uninstall** — extension is installed. Click to remove it.
- **Disable** — extension is installed and turned on. Click to turn it off without removing it. Useful when an extension is misbehaving but you'll want it back.

Always look at the **publisher** under the extension name before installing. There are sometimes multiple extensions with the same name. Trusted publishers for this course: **Microsoft**, and the language/tool's official organization (e.g. `dbaeumer` for ESLint, `ms-python` for Python).

Course-recommended extensions

For this course you will install these. We'll install some now and some in later lessons of this module:

Extension	Publisher	Lesson	Purpose
Python	Microsoft (`ms-python`)	this lesson	Language support for Python
Docker	Microsoft (`ms-azuretools`)	lesson 04	Sidebar for inspecting containers
WSL	Microsoft (`ms-vscode-remote`)	lesson 05 (Windows only)	Connect VS Code to WSL
Remote - SSH	Microsoft (`ms-vscode-remote`)	lesson 06	Connect VS Code to a remote machine
Continue	Continue.dev (`Continue.continue`)	Module 05	Free AI assistant — chat, inline edit, autocomplete

> Don't install random extensions today. Stick to the list above. **Continue.dev** is our free AI extension, but we don't install it until Module 05 — leave it for now. We'll revisit extensions for actual work in later modules.

Try it yourself: install Python

1. Open the extensions side bar (`Ctrl+Shift+X`).
2. In the search box at the top, type **Python**.
3. The first result should be "Python" by Microsoft. Verify the publisher says **Microsoft** before clicking.
4. Click **Install**.
5. After a few seconds, the button changes to a gear/disable icon — that means it installed.
6. Scroll the side bar down to the **INSTALLED** section. You should see "Python" listed there.

Disabling and uninstalling

You won't do this today, but for reference:

- To turn an extension off temporarily: open the extensions side bar, click the extension, click the **Disable** dropdown next to the gear.
- To remove it: same place, choose **Uninstall**.
- After enable/disable, VS Code sometimes asks to reload the window. Click "Reload."

Checkpoint

You're done when:

- [ ] You opened the extensions side bar with `Ctrl+Shift+X`.
- [ ] You searched for and installed the **Python** extension by **Microsoft**.
- [ ] "Python" appears under the **INSTALLED** section in the extensions side bar.
- [ ] You know the difference between Install, Disable, and Uninstall.

If you accidentally installed a different "Python" extension (wrong publisher), uninstall it and install the Microsoft one.

Lab

Objective: Install an extension from the marketplace and verify its publisher before installing.

Task: Work through this lesson — open the extensions side bar (`Ctrl+Shift+X`), search for **Python**, verify the publisher is **Microsoft**, and install it.

Submit: One screenshot of the extensions side bar showing "Python" under the **INSTALLED** section, with the publisher (Microsoft) visible.

Pass criteria:

- [ ] "Python" appears under the INSTALLED section of the extensions side bar.
- [ ] The publisher shown is Microsoft (`ms-python`).
- [ ] No extra extensions were installed beyond the course list.

Unlocks: 04-docker-extension.md

> **Lab — do it for real.** Complete every step above on your own machine — type the commands, don't copy-paste. **Submit:** a screenshot or pasted output of the *Checkpoint* section (or this lesson's final verification steps) Keep it. If you cannot submit in D2L, bring it to class or email it to your instructor. Completing the lab unlocks the next lesson.

Learning objective: Install the Docker extension by Microsoft, locate its whale icon and sidebar sections, and explain why the sidebar shows 'Failed to connect' before Docker is installed.

Lesson 04: The Docker Extension

Time: ~10 min

What is Docker, in one paragraph?

Docker is a tool for packaging an application together with everything it needs to run (its operating system files, libraries, configs) into a single bundle called a **container**. You can ship that container to any computer that has Docker installed and it will run the same way. Containers are like very lightweight virtual machines.

You don't need to fully understand Docker today. You just need to know:

- Docker is installed in **Module 02**, not now.
- VS Code has a **Docker extension** (Microsoft's "Container Tools") that gives you a sidebar for looking at your containers, images, etc. We're installing that extension now so it's ready when you need it.

Important: what you'll see vs what works

Because Docker isn't installed yet, the Docker sidebar will say **"Failed to connect"** or just be empty when you open it. **This is correct.** You haven't done anything wrong. The extension is fine — it just has nothing to talk to until Module 02.

Your instructor will show you what the sidebar **will** look like after Module 02 (with their own Docker running, or a screenshot).

Try it yourself: install the Docker extension

1. Open the extensions side bar (`Ctrl+Shift+X`).
2. Search for **Docker**.
3. The one you want is published by **Microsoft** (publisher id `ms-azuretools.vscode-docker`). It usually has the most installs.
4. Click **Install**.
5. After install, look at the activity bar on the left edge of the window. There's a new icon: a whale with a stack on its back. That's the Docker side bar.

Tour the Docker side bar

1. Click the Docker whale icon in the activity bar.

2. The side bar shows several sections:

- **CONTAINERS** — running and stopped containers.
- **IMAGES** — the read-only blueprints that containers are built from.
- **NETWORKS** — virtual networks Docker has created.
- **VOLUMES** — persistent storage attached to containers.
- **REGISTRIES** — places you can pull or push images to (Docker Hub, etc.).

1. All of these will show "Failed to connect" or "No items found." Again — expected.

You don't need to understand what every section does today. Just remember they exist. Once Module 02 is done, this sidebar will fill up and you'll come back to it.

Checkpoint

You're done when:

- [ ] You installed the **Docker** extension by **Microsoft** (`ms-azuretools.vscode-docker`).
- [ ] The Docker whale icon is in your activity bar on the left.
- [ ] You clicked it and saw the CONTAINERS / IMAGES / NETWORKS / VOLUMES / REGISTRIES sections (even if they're empty or showing an error).
- [ ] You can explain in one sentence why the sidebar is empty right now.

Lab

Objective: Install the Docker extension so it's ready for Module 02, and understand why its sidebar is empty today.

Task: Work through this lesson — install the **Docker** extension by Microsoft (`ms-azuretools.vscode-docker`), then click the whale icon and tour the sidebar sections.

Submit: One screenshot showing the Docker whale icon in your activity bar with the Docker side bar open (its "Failed to connect" or empty sections visible), plus one sentence in the submission comment explaining why the sidebar is empty right now.

Pass criteria:

- [ ] The whale icon appears in the activity bar.
- [ ] The Docker side bar sections (CONTAINERS / IMAGES / NETWORKS / VOLUMES / REGISTRIES) are visible, empty or showing an error.
- [ ] Your one-sentence comment says the sidebar is empty because Docker itself isn't installed until Module 02.

Unlocks: 05-wsl-connection.md

Lab — do it for real. Complete every step above on your own machine — type the commands, don't copy-paste. **Submit:** a screenshot or pasted output of the *Checkpoint* section (or this lesson's final verification steps) Keep it. If you cannot submit in D2L, bring it to class or email it to your instructor. Completing the lab unlocks the next lesson.